PingSheng Kao

Santa Clara, California | +1 734 800 6888 | samk840125@gmail.com | linkedin.com/in/pingshengkao

SUMMARY

ML Data Scientist with 5+ years of experience evaluating and prototyping multimodal conversational agents and production ML systems. Builds simulated users, agent-as-judge workflows, behavioral scenarios, expert-aligned benchmarks, and model-repair experiments to uncover and improve complex failure modes.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Built a simulated-user agent, agent-as-judge workflow, targeted scenario generators, and user-centered metrics to stress-test an open-ended multimodal conversational agent and surface visual-memory, sycophancy, intent-misalignment, and false inferred-state-update failures.
- Deployed and scaled LLM-as-judge and agent-as-judge evaluation on an existing internal batch platform, evaluating factual consistency and tool-call behavior, debugging throughput and rate-limit issues, and surfacing results in dashboards.
- Developed scene-graph-based evaluation for free-form image descriptions, measuring OCR, counting, attribute, and spatial-reasoning failures in multimodal-model outputs.
- Built an auditable, reference-based multi-stage LLM evaluator and expert-aligned data workflow for a multimodal health-related AI product, reaching 90% alignment with experts on a 40-example gold set.
- Trained and fine-tuned vision models on VLM-discovered and relighting-derived failure slices and benchmarked targeted repair against alternative slice-discovery approaches.
- Developed embedding-driven failure discovery for Vision Pro hand tracking using similarity search, clustering, OOD, and position-level analysis; contributed to final quality metrics below 2% missing pinch and 5 false positives per hour.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments using LLMs and VLMs to generate and verify grounded failure hypotheses for fine-grained, interpretable error-slice discovery.
- Experiment contribution only; credited as Sam Kao and not a listed author.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming: Python, SQL, C++, JavaScript
Agents & Multimodal: User Simulation, Agent-as-Judge, Tool-Call Evaluation, VLM/MLLM Evaluation, Scene Graphs
ML & Research: PyTorch, Vision-Model Fine-Tuning, Experiment Design, Embeddings, Clustering, OOD Analysis
Evaluation: LLM-as-Judge, Behavioral Failure Analysis, Ground-Truth Design, Human Evaluation, DSPy